
Algorithm 1: BEAMSEARCH(s, q, k, L)

Input: Graph G , start node s , query q , result size k , search list size $L \geq k$

Output: Result set $\mathcal{L}$ (k -approx nearest neighbors) and visited set $\mathcal{V}$

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1  $\mathcal{L} \leftarrow \{s\}, \quad \mathcal{E} \leftarrow \emptyset, \quad \mathcal{V} \leftarrow \emptyset$ 
2 while  $\mathcal{L} \setminus \mathcal{E} \neq \emptyset$  do
3    $p^\circ \leftarrow \arg \min_{x \in \mathcal{L} \setminus \mathcal{E}} D(x, q)$ 
4    $\mathcal{L} \leftarrow \mathcal{L} \cup (N_{\text{out}}(p^\circ) \setminus \mathcal{V}), \quad \mathcal{E} \leftarrow \mathcal{E} \cup \{p^\circ\}$ 
5   if  $|\mathcal{L}| > L$  then
6     | update  $\mathcal{L}$  to retain the  $L$  closest points to  $q$ 
7   end
8    $\mathcal{V} \leftarrow \mathcal{V} \cup N_{\text{out}}(p^\circ)$ 
9 end
10 return the  $k$  closest points from  $\mathcal{V}$ ;  $\mathcal{V}$ 
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